Randomized Algorithms

— P, NP, RP, PP, ZPP, BPP, ...

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Outline

1. RAMs & Turing Machines
2. Complexity Classes
   - Deterministic & Non-Deterministic Classes
   - Space Complexity Classes
   - Reduction & Completeness
   - Randomized Complexity Classes
3. Transformation of Probability Distributions

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RAM (Random Access Machine)

- RAM is a model of computation used when describing and analyzing algorithms.
- A machine can perform operations involving registers and main memory.
- The *unit-cost* RAM: each instruction can be performed in one time step.
  - Too powerful; no known polynomial time simulation of this type of model by Turing machines.
- The *log-cost* RAM: each instruction requires time proportional to the logarithm of the size of its operands.

Turing Machine

A physical Turing machine (with finite amount of tape).

Deterministic Turing Machine

A deterministic Turing machine is a quadruple $M = (S, \Sigma, \delta, s)$.

- S: a finite set of states (s: the initial state)
- Σ: A finite set of symbols (including special symbols BLANK and FIRST).
- δ: the transition function.
  - $S \times \Sigma \to (S \cup \{\text{HALT, YES, NO}\}) \times \Sigma \times \{\leftarrow, \rightarrow, \text{STAY}\}$.
  - HALT, YES, NO: The three halting states not in S.

Turing Machine (Input & Tape)

- The input to the TM: written on a tape.
- The TM, as an algorithm, may read from and write on this tape.
- Assume that HALT, YES, NO as well as the symbols $\leftarrow, \rightarrow$, and STAY are not in $S \cup \Sigma$.
- The TM begins in the initial state s with its cursor at the first symbol FIRST of input x.
- The input is a string of $(\Sigma \setminus \{\text{BLANK, FIRST}\})^*$.
  - The left-most BLANK on the tape: the end of the input string.

Turing Machine (Transition)

- The transition function δ: can be thought as a *program*.
- In each step, the TM reads the symbol α pointed by the cursor;
- Based on α and the current state, choose:
  - a next state;
  - a symbol β to be overwritten on α;
  - a cursor motion direction from $\{\leftarrow, \rightarrow, \text{STAY}\}$.
- The cursor never falls off the left end of the input: FIRST.
- The BLANK symbol can be overwritten.

Turing Machine (Acceptance & Reject)

- The TM has accepted the input x: if the TM halts in the YES state.
- The TM has rejected the input x: if the TM halts in the NO state.
- State HALT: for the computation of functions whose range is not Boolean (output of the function is written on the tape).

Physical Turing Machine M (Finite Tape)

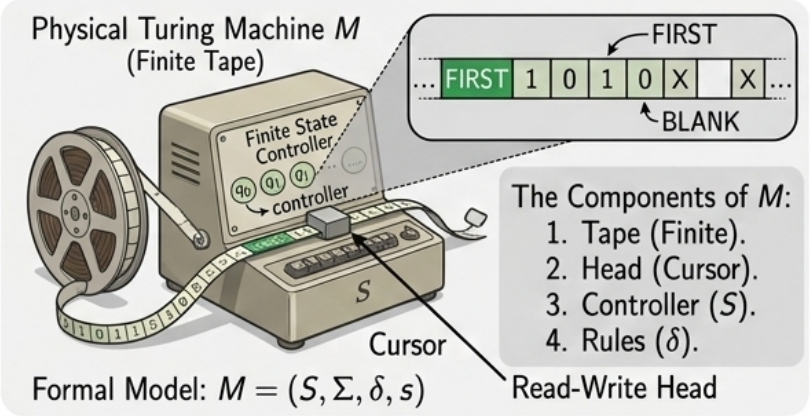

Figure 1: Anatomy of a Deterministic Turing Machine (DTM)

δ: Transition and Program View

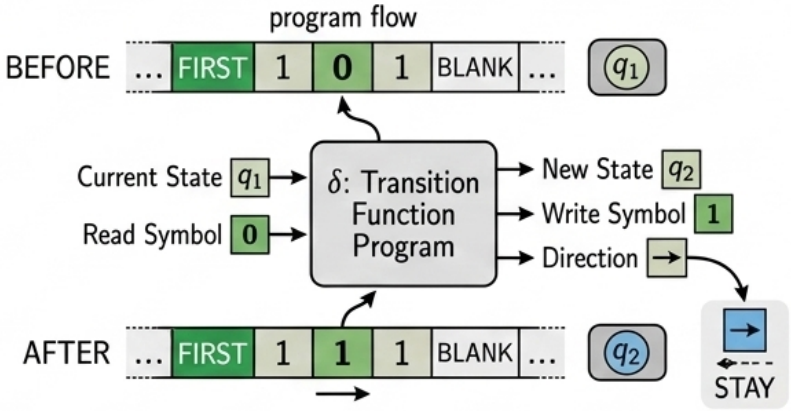

Figure 3: Transition and Program View

The Start Configuration and Tape Structure

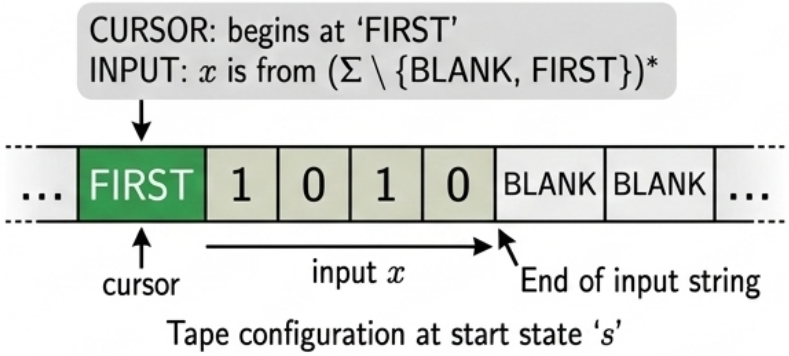

Figure 2: The Start Configuration and Tape Structure

Fig 4: Halting Conditions and Results

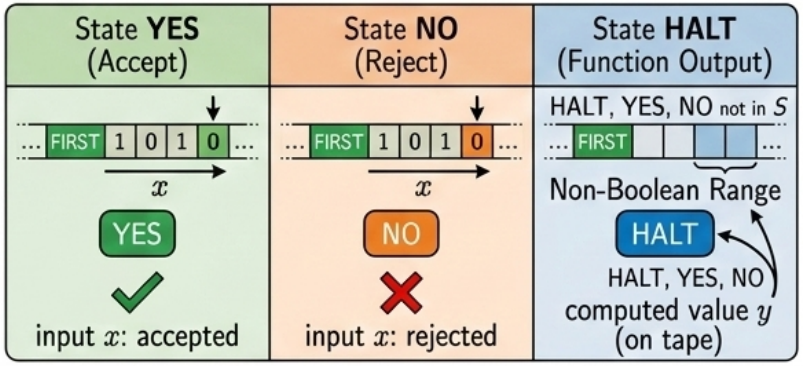

Figure 4: Halting Conditions and Results

Probabilistic Turing Machine

A probabilistic Turing machine is a Turing machine augmented with the ability to generate an unbiased coin flip in one step.

- This corresponds to a randomized algorithm.

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SAT

An instance of satisfiability (SAT):

$$(x_1 \wedge \neg x_2 \wedge x_4) \vee (\neg x_3 \wedge \neg x_4 \wedge x_5) \vee (\neg x_1 \wedge x_2 \wedge x_4 \wedge \neg x_5)$$

- $x_1, x_2, \ldots$: variables
- $\neg x_1, x_2$: literals
- $(\cdots)$: clauses

Language Recognition Problems

Language Recognition Problems

Any decision problem can be treated as a language recognition problem.

- Σ^*: the set of all possible strings over Σ.
- $|S|$: length of string s.

A language $L \subseteq \Sigma^*$ is any collection of strings over Σ.

A Language Recognition Problem

Decide whether a given string $x \in \Sigma^*$ belongs to L.

Complexity Class

A collection of languages all of whose recognition problems can be solved under prescribed bounds on the computational resources.

P & *NP*

P

The class ***P*** consists of all languages L which has a polynomial time algorithm A s.t. for any input $x \in \Sigma^*$,

- $x \in L \Rightarrow A(x)$ accepts;
- $x \notin L \Rightarrow A(x)$ rejects.

NP

The class ***NP*** consists of all languages L which has a polynomial time algorithm A s.t. for any input $x \in \Sigma^*$,

- $x \in L \Rightarrow \exists y \in \Sigma^*$, $A(x, y)$ accepts for $|y| \leq \text{poly}(|x|)$;
- $x \notin L \Rightarrow \forall y \in \Sigma^*$, $A(x, y)$ rejects for $|y| \leq \text{poly}(|x|)$.

For example, given an instance of satisfiability (SAT):

$$\mathbf{x} = (x_1 \wedge \neg x_2 \wedge x_4) \vee (\neg x_3 \wedge \neg x_4 \wedge x_5) \vee (\neg x_1 \wedge x_2 \wedge x_4 \wedge \neg x_5)$$

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Suppose we have the "proof" / "certificate" $\mathbf{y}$ as

$$x_1 : \text{True},\ x_2 : \text{False},\ x_3 : \text{True},\ x_4 : \text{True},\ x_5 : \text{True}$$

For example, given an instance of satisfiability (SAT):

$$\mathbf{x} = (x_1 \wedge \neg x_2 \wedge x_4) \vee (\neg x_3 \wedge \neg x_4 \wedge x_5) \vee (\neg x_1 \wedge x_2 \wedge x_4 \wedge \neg x_5)$$

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$$x_1 : \text{True},\ x_2 : \text{False},\ x_3 : \text{True},\ x_4 : \text{True},\ x_5 : \text{True}$$

We can check that $(\mathbf{x}, \mathbf{y}) \in L$ (encoded as string of $O(\log n)$ space in the tape) in polynomial time, where L denote the set of all satisfiable formula.

A Useful, Alternative Viewpoint

The class $\boldsymbol{P}$ consists of all language L such that for any $x \in L$, a proof of $x \in L$ (represented by the string y) can be found and verified in polynomial time.

The class $\boldsymbol{NP}$ consists of all language L such that for any $x \in L$, a proof of $x \in L$ (represented by the string y) can be verified in polynomial time.

Obviously,

$$\boldsymbol{P} \quad \boldsymbol{NP}.$$

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Obviously,

$$\boldsymbol{P} \subseteq \boldsymbol{NP}.$$

Complementary Classes

For any complexity class $\mathcal{C}$, the complementary class co-$\mathcal{C}$ is the set of languages whose complement is in $\mathcal{C}$. That is,

$$\text{co-}\mathcal{C} = \{L \mid \bar{L} \in \mathcal{C}\}.$$

Examples: co-P & co-NP

co-P

The class co-$\boldsymbol{P}$ consists of all languages L which has a polynomial time algorithm A s.t. for any input $x \in \Sigma^*$,

- $x \notin L \Rightarrow A(x)$ accepts;
- $x \in L \Rightarrow A(x)$ rejects.

co-NP

The class co-$\boldsymbol{NP}$ consists of all languages L which has a polynomial time algorithm A s.t. for any input $x \in \Sigma^*$,

- $x \notin L \Rightarrow \exists y \in \Sigma^*$, $A(x, y)$ accepts for $|y| \leq \text{poly}(|x|)$;
- $x \in L \Rightarrow \forall y \in \Sigma^*$, $A(x, y)$ rejects for $|y| \leq \text{poly}(|x|)$.

P & co-P

P

The class $\boldsymbol{P}$ consists of all languages L which has a polynomial time algorithm A s.t. for any input $x \in \Sigma^*$,

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Questions: $\boldsymbol{P} =$ co-$\boldsymbol{P}$?

NP & co-*NP*

NP

The class ***NP*** consists of all languages L which has a polynomial time algorithm A s.t. for any input $x \in \Sigma^*$,

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- $x \notin L \Rightarrow \forall y \in \Sigma^*$, $A(x, y)$ rejects for $|y| \leq \text{poly}(|x|)$.

co-*NP*

The class co-***NP*** consists of all languages L which has a polynomial time algorithm A s.t. for any input $x \in \Sigma^*$,

- $x \notin L \Rightarrow \exists y \in \Sigma^*$, $A(x, y)$ accepts for $|y| \leq \text{poly}(|x|)$;
- $x \in L \Rightarrow \forall y \in \Sigma^*$, $A(x, y)$ rejects for $|y| \leq \text{poly}(|x|)$.

Open Questions: $\boldsymbol{P} = \boldsymbol{NP} \cap \text{co-}\boldsymbol{NP}$? $\quad \boldsymbol{NP} = \text{co-}\boldsymbol{NP}$?

Example: SAT & UNSAT

- SAT $\in$ ***NP***: a satisfying assignment is an easy certificate.
- UNSAT $\in$ co-***NP***: to show a formula is unsatisfiable, you need a certificate that **no** assignment works.

Similarly, . . .

EXP & NEXP

EXP

The class ***EXP*** consists of all languages L which has an exponential time algorithm A s.t. for any input $x \in \Sigma^*$,

- $x \in L \Rightarrow A(x)$ accepts;
- $x \notin L \Rightarrow A(x)$ rejects.

NEXP

The class ***NEXP*** consists of all languages L which has an exponential time algorithm A s.t. for any input $x \in \Sigma^*$,

- $x \in L \Rightarrow \exists y \in \Sigma^*$, $A(x, y)$ accepts for $|y| \leq \text{poly}(|x|)$;
- $x \notin L \Rightarrow \forall y \in \Sigma^*$, $A(x, y)$ rejects for $|y| \leq \text{poly}(|x|)$.

A Useful, Alternative Viewpoint

The class ***EXP*** consists of all language L such that for any $x \in L$, a proof of $x \in L$ (represented by the string y) can be found and verified in exponential time.

The class ***NEXP*** consists of all language L such that for any $x \in L$, a proof of $x \in L$ (represented by the string y) can be verified in exponential time.

Obviously,

$$\boldsymbol{EXP} \subseteq \boldsymbol{NEXP}.$$

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Space

- The space used by a TM: the number of distinct positions on the tape that are scanned during an execution.
  - For RAMs, its the number of words of memory required by an algorithm.
- ***PSPACE*** and ***NPSPACE***: resembles the settings of ***P*** and ***NP*** but requiring polynomial space.
- A ***PSPACE*** algorithm may run for super-polynomial time (e.g., $2^{\text{poly}(n)}$).
- Known results: ***PSPACE*** = ***NPSPACE***, ***PSPACE*** = co-***PSPACE***.
  - Savitch's theorem: a deterministic Turing machine can simulate a nondeterministic Turing machine without needing much more space.

Intuition of Savitch's Theorem (Supplementary)

- A nondeterministic TM may "branch" into many possible next moves, while a deterministic TM follows only one computation path.
- At first glance, simulating all branches seems to require tracking exponentially many possibilities at once.
- Savitch's theorem: *NOT* necessary for *space*: a deterministic TM can explore those possibilities reusing the same memory.

Savitch's Theorem [1970]

For every space-constructible function $s(n) \geq \log n$,

$$\mathbf{NSPACE}(s(n)) \subseteq \mathbf{DSPACE}(s(n)^2).$$

- If a nondeterministic TM uses $s(n)$ space, then a deterministic TM can simulate it using $\leq c \cdot s(n)^2$ space for some constant c.
  - **Key consequence:** $\mathbf{PSPACE} = \mathbf{NPSPACE}$.

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Reduction

Polynomial Reduction

A polynomial reduction from a language $L_1 \subseteq \Sigma^*$ to a language $L_2 \subseteq \Sigma^*$ is a function $f : \Sigma^* \mapsto \Sigma^*$ such that

- $\exists$ a polynomial time algorithm that computes f
- $\forall x \in \Sigma^*$, $x \in L_1$ if and only if $f(x) \in L_2$.

Completeness

NP-hard

A language L is NP-hard if, for all $L' \in NP$, there is a polynomial reduction from L' to L.

NP-complete

A language L is NP-complete if it is in NP and is NP-hard.

The first NP-complete problem: SAT (Cook-Levin Theorem (1971)).

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RP

RP

The class ***RP*** (i.e., Randomized Polynomial time) consists of all languages L that have a randomized algorithm A which runs in worst-case polynomial time such that for any input $x \in \Sigma^*$:

- $x \in L \Rightarrow \Pr[A(x) \text{ accepts}] \geq \frac{1}{2}$.
- $x \notin L \Rightarrow \Pr[A(x) \text{ accepts}] = 0$.

- Err only when $x \in L$. $\Rightarrow$ one-sided error.

co-*RP*

co-*RP*

The class co-***RP*** (i.e., complement Randomized Polynomial time) consists of all languages L that have a randomized algorithm A which runs in worst-case polynomial time such that for any input $x \in \Sigma^*$:

- $x \notin L \Rightarrow \Pr[A(x) \text{ accepts}] \geq \frac{1}{2}$.
- $x \in L \Rightarrow \Pr[A(x) \text{ accepts}] = 0$.

- Err only when $x \notin L$. $\Rightarrow$ one-sided error.

Example of *RP*

Min-Cut$_{\leq k}$

$$\text{Min-Cut}_{\leq k} = \{\, (G, k) \mid G \text{ has a cut of size at most } k \,\}.$$

Exercise

Assume that we have the following class:

RP'

The class $\boldsymbol{RP'}$ consists of all languages L that have a randomized algorithm A which runs in worst-case polynomial time such that for any input $x \in \Sigma^*$:

- $x \in L \Rightarrow \Pr[A(x) \text{ accepts}] \geq \frac{1}{n^2}$.
- $x \notin L \Rightarrow \Pr[A(x) \text{ accepts}] = 0$.

Prove that $\boldsymbol{RP'} = \boldsymbol{RP}$.

$RP \cap$ co-RP

ZPP (Zero-error Probabilistic Polynomial time

The class ZPP is the class of languages that have Las Vegas algorithms running in expected polynomial time.

Why *ZPP*?

- Suppose we have a language $L \in \boldsymbol{RP} \cap \text{co-}\boldsymbol{RP}$.
- L can be recognized by an ***RP*** algorithm A and a co-***RP*** algorithm B.

A Las Vegas algorithm

Given the input x, perform the following procedure in iterations.

1. If $A(x)$ accepts, then x must be a YES-instance;
2. Otherwise, if $B(x)$ rejects, then x must be a NO-instance.
3. If neither of above occurs, continue to next iteration.

- The expected number of iterations is bounded!

PP

PP

The class ***PP*** (i.e., Probabilistic Polynomial time) consists of all languages L that have a randomized algorithm A running in worst-case polynomial time such that for any input $x \in \Sigma^*$:

- $x \in L \Rightarrow \Pr[A(x) \text{ accepts}] > \frac{1}{2}$.
- $x \notin L \Rightarrow \Pr[A(x) \text{ accepts}] < \frac{1}{2}$.

PP

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- $x \in L \Rightarrow \Pr[A(x) \text{ accepts}] > \frac{1}{2}$.
- $x \notin L \Rightarrow \Pr[A(x) \text{ accepts}] < \frac{1}{2}$.

- To reduce the error probability of a two-sided error algorithm, we can perform several independent iterations on the same input.
- Output the majority answer of these iterations.

*BPP*BPP

BPP

The class ***BPP*** (i.e., Bounded-error Probabilistic Polynomial time) consists of all languages L that have a randomized algorithm A running in worst-case polynomial time such that for any input $x \in \Sigma^*$, $n := |x|$:

- $x \in L \Rightarrow \Pr[A(x) \text{ accepts}] \geq \frac{3}{4}$.
- $x \notin L \Rightarrow \Pr[A(x) \text{ accepts}] \leq \frac{1}{4}$.

BPPBPP

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- $x \in L \Rightarrow \Pr[A(x) \text{ accepts}] \geq \frac{3}{4}$.
- $x \notin L \Rightarrow \Pr[A(x) \text{ accepts}] \leq \frac{1}{4}$.

- To reduce the error probability of a two-sided error algorithm, we can perform several independent iterations on the same input.
- Output the majority answer of these iterations.
- ***BPP*** $\subseteq$ ***PP***.

Quesion: *BPP*?

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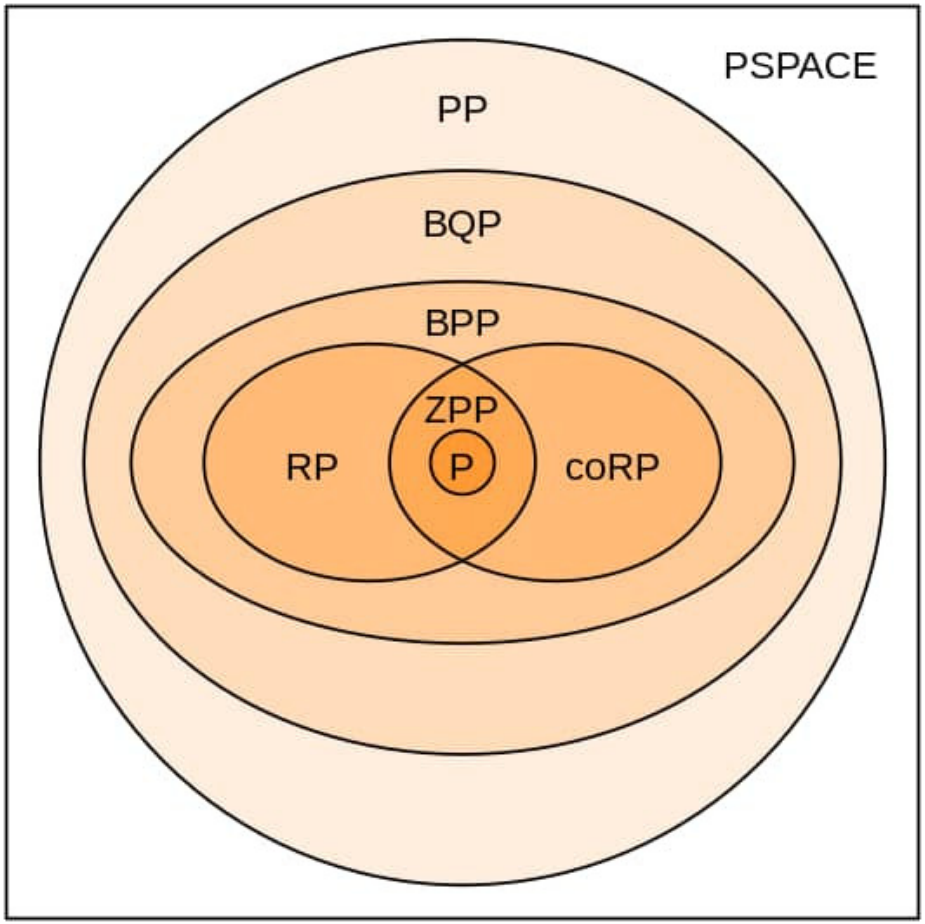
- $x \in L \Rightarrow \Pr[A(x) \text{ accepts}] \geq \frac{1}{2} + \frac{1}{\text{poly}(n)}$.
- $x \notin L \Rightarrow \Pr[A(x) \text{ accepts}] \leq \frac{1}{2} - \frac{1}{\text{poly}(n)}$.

- To reduce the error probability of a two-sided error algorithm, we can perform several independent iterations on the same input.
- Output the majority answer of these iterations.

Randomized Complexity Classes

Source: Wikipedia



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p-coin & Transforming a Probability Distribution

p-coin

A coin is called a p-coin if it shows HEAD after one coin-flipping with probability p.

Probability Distribution Transformations

A function that transforms a p-coin to get a q-coin, for $0 < p, q < 1$, is called a p-to-q transformation.

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Easy cases:

- p-to-0 and p-to-1..

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- p-to-$(1 - p)$
- p-to-p^2.

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Probability Distribution Transformations

A function that transforms a p-coin to get a q-coin, for $0 < p, q < 1$, is called a p-to-q transformation.

Easy cases:

- p-to-0 and p-to-1..
- p-to-$(1-p)$
- p-to-p^2.
- p-to-$(\binom{n}{k} p^k (1-p)^{n-k})$, for $n \in \mathbb{N}$ and $k \in \{0, 1, \ldots, n\}$.

Exercise

Given a p-coin, where $0 < p < 1$.

- Repeat the following steps until it returns YES or NO.
  1. flip the p-coin twice.
  2. if the results are HEAD-TAIL, return YES;
  3. else if the results are TAIL-HEAD, return NO;
  4. otherwise, continue to next iteration

- Please prove that the above procedure is a p-to-$\frac{1}{2}$ transformation (i.e., deriving a fair coin).
- Please compute the expected number of coin-flips of the above procedure.

Concatenation Rule

Concatenation Rule

Given a p-coin and a q-coin, we can derive a pq-coin as follows.

1. First, simulate the p-coin. If it is TAIL, output TAIL
2. Otherwise, simulate the q-coin and output the outcome.

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Given a p-coin and a q-coin, we can derive a pq-coin as follows.

1. First, simulate the p-coin. If it is TAIL, output TAIL
2. Otherwise, simulate the q-coin and output the outcome.

- By the Concatenation Rule & the exercise, we can derive a $p/2$-coin when we are given a p-coin.

Reference

- Fabian Frei and Peter Rossmanith: Transformations of probability distributions. [Theoretical Computer Science 2024].

Discussions